

Study Committee C6

Active Distribution Systems and Distributed Energy Resources

PS1_11391_2026

Alternate Framing: Predictive Control and Economic Assessment of Grid-Forming BESS for Flexibility Provision

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Motivation

The rapid increase of inverter-based resources requires Battery Energy Storage Systems (BESS) capable of providing both operational flexibility and economic value. Although energy arbitrage is one of the main applications of BESS, its long-term profitability strongly depends on dispatch strategy, battery efficiency and degradation. Most existing studies neglect the interaction between hourly operation and long-term ageing. This work addresses this gap through an integrated techno-economic framework.

Method/Approach

An hourly simulation framework was developed in MATLAB, combining:

- Hourly SOC evolution
- SOH degradation (calendar + cycling)
- Heuristic and MILP dispatch
- Economic evaluation (NPV, ROI and cash flow)
- Break-even CAPEX analysis

Two BESS technologies with identical ratings but different round-trip efficiencies were evaluated over a 10-year horizon.

Problem Characterization

Most arbitrage studies assume:

- Constant battery efficiency
 - Fixed lifetime
 - Simplified degradation
 - Rule-based dispatch
- These assumptions generally overestimate revenues and underestimate degradation. This work explicitly couples operational flexibility, degradation and economics.

Study Case and Modelling

- 10 MW / 20 MWh Grid-Forming BESS
- Energy-only market
- Hourly synthetic electricity prices
- 10-year simulation
- Calendar + cycling degradation
- Identical CAPEX and OPEX
- Different battery efficiencies (BESS-A, BESS-B).

Simulation Scenarios

Two dispatch strategies were investigated:

Scenario 1

Threshold-based heuristic dispatch using fixed charging and discharging prices

Scenario 2

MILP optimal dispatch maximizing annual arbitrage revenue while considering:

- SOC dynamics
- Power limits
- Binary charging/discharging constraints
- Terminal SOC.

Results

Figure 1 shows the higher efficiency allows BESS-A to sustain discharge during peak-price periods

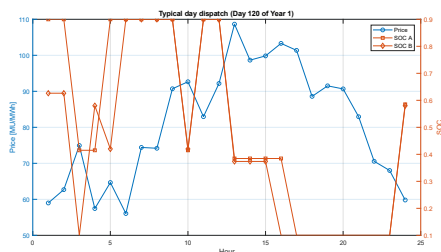


Figure 1 – Typical daily dispatch via MILP

Figure 2-a and 2-b illustrate that the higher-efficiency BESS-A consistently achieves greater energy throughput and higher annual cash flows.

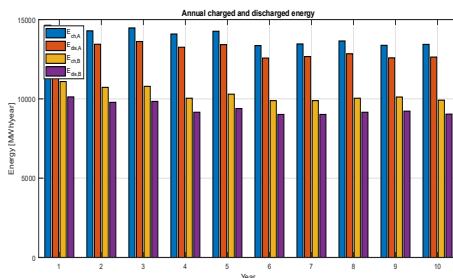


Figure 2-a) Annual charged/discharged energy

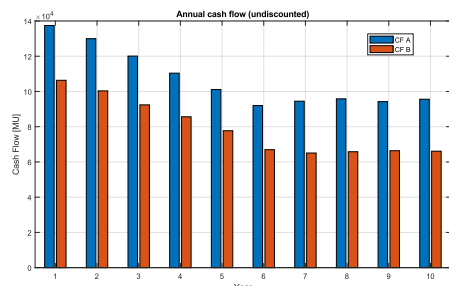


Figure 2-b) Annual cash flow (undiscounted)

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Figure 3-a and 3-b demonstrate that the economic advantage of higher battery efficiency accumulates throughout the project lifetime.

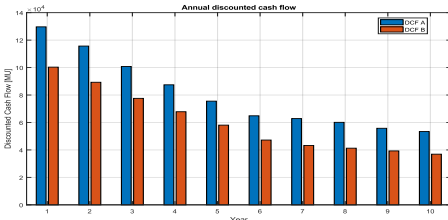


Figure 3-a) Annual discounted cash flow

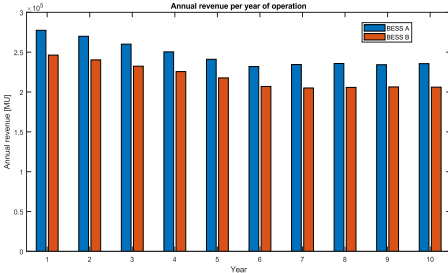


Figure 3-b) Annual revenue per year of operation

Figure 4 highlights the trade-off between operational flexibility and battery degradation. Higher throughput increases economic value but accelerates SOH degradation.

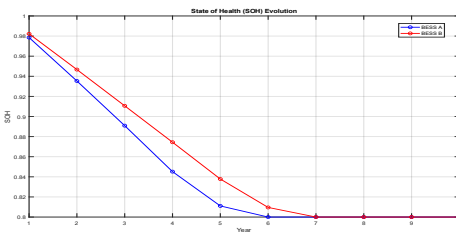


Figure 4) State of health evolution in 10 years

Table 1 shows that MILP increases operating profit and energy throughput through optimal scheduling, but also accelerates battery degradation. Although NPV improves, energy arbitrage alone remains insufficient under the assumed market conditions.

Table 1) Method comparison (Heuristic vs MILP)

Metric	A (Heur.)	A (MILP)	B (Heur.)	B (MILP)
NPV [MU]	-5801897	-5193919	-5885210	-5398967
ROI [%]	-96.70	-86.57	-98.09	-89.98
Profit (undisc.) [MU]	253012	1071070	141278	792686
Arbitrage revenue [MU]	1653012	2471070	1541278	2192686
Total charged energy [MWh]	42273.9	139038.1	42589.1	102795.4
Total discharged energy [MWh]	39065.5	130822.5	38143.0	93751.0
Total equivalent cycles	2862.9	9887.6	2854.3	7147.6
Final SOH	0.800	0.800	0.800	0.800
Average SOH	0.888	0.846	0.888	0.856
Revenue per discharged MWh [MU/MWh]	42.31	18.89	40.41	23.39

Tables 2 and 3 show that MILP enhances the economic value of BESS-A. As revenues come solely from energy arbitrage, the gains are driven by operational flexibility and battery efficiency.

Table 2) A vs B difference by method (robust)

Indicator	Heuristic	MILP
ANPV (A-B) [MU]	83313	205048
ANPV/CAPEX [%]	1.39	3.42

Table 3) Revenue breakdown (value stacking)

Metric	BESS A	BESS B
Total revenue [MU]	2471070	2192686
Arbitrage revenue [MU]	2471070 (100.0%)	2192686 (100.0%)

Analysis of Results

- MILP significantly increases energy throughput.
- Operating profit is considerably higher than with heuristic dispatch
- Greater flexibility produces more equivalent cycles
- Higher throughput accelerates battery degradation
- Dispatch optimisation cannot eliminate efficiency losses.

Implications

The proposed methodology enables:

- Technology benchmarking
- CAPEX sensitivity analysis
- Investment decision support
- Evaluation of operational flexibility
- Long-term economic assessment, including degradation.

The framework can also be extended to include ancillary services and capacity markets.

Conclusions

Operational flexibility creates economic value, but it also accelerates battery degradation. Therefore, realistic techno-economic assessments must jointly model dispatch optimisation, battery ageing and long-term financial performance.

The highlights are:

- MILP dispatch substantially improves arbitrage revenues.
- Increased flexibility leads to higher battery utilisation and faster ageing.
- Battery efficiency has a significant long-term economic impact.
- Higher-efficiency technologies justify a measurable CAPEX premium.
- The proposed SOC-SOH-MILP framework provides a realistic methodology for techno-economic assessment of Grid-Forming BESS.